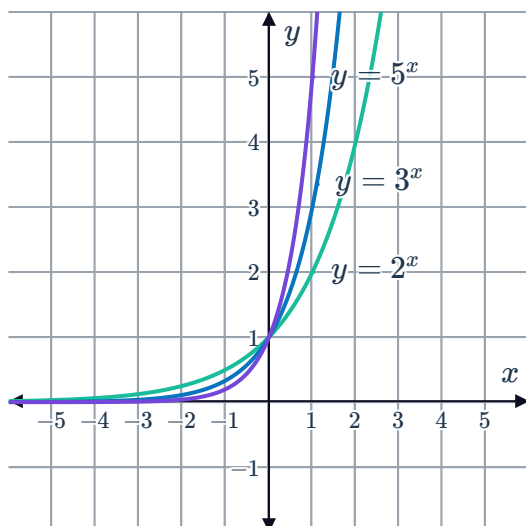


Exponential graphs and transformations



1

a



b

- i Yes ii Yes iii No iv No

c Shared y -intercept of $(0, 1)$

2

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y	$-\frac{1}{32}$	$-\frac{1}{16}$	$-\frac{1}{8}$	$-\frac{1}{4}$	$-\frac{1}{2}$	-1	-2	-4	-8	-16	-32

b No. For $a > 0$ the expression a^x is greater than zero for all real x . Hence, $-(a^x)$ will be less than or equal to zero.

c Decreasing

d As x increases, the function decreases at a faster and faster rate.

e All real x .

f $y < 0$

3

a -243

b -3

c $-\frac{1}{3}$

d 4

4

a 1

b 1

c -1

d -1

5

a Decreasing

b 0

c $-\infty$

d $(0, -2.5)$

6

a Decreasing

b Increasing

c Decreasing

d Increasing

e Increasing

f Increasing

g Decreasing

h Decreasing

i Decreasing

j Increasing

k Increasing

l Increasing



7 B

8 $y = 3 \times 2^x$

9

a 1

b -1

c $y = 0$

d $y = -2$

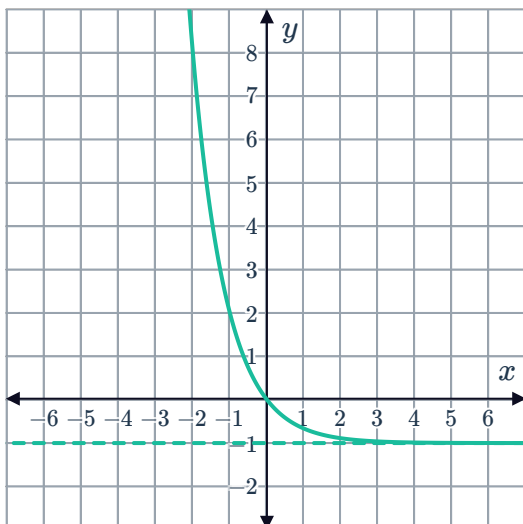
10

a

i 0

ii $y = -1$

iii

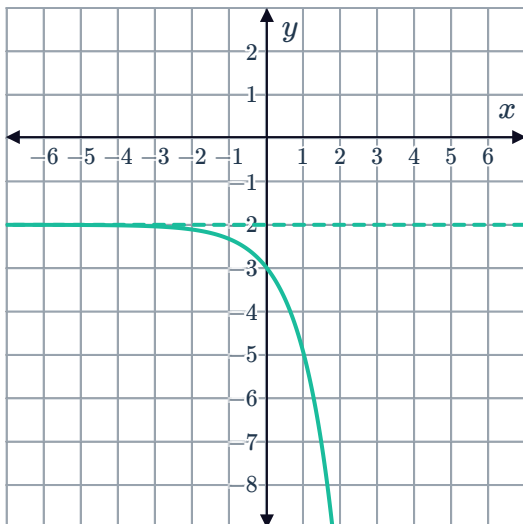


b

i -3

ii $y = -2$

iii





11



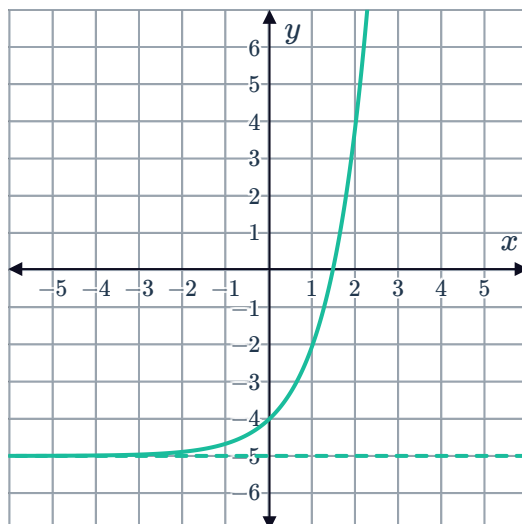
a

i -4

iii $y > -5$

ii All real x .

iv



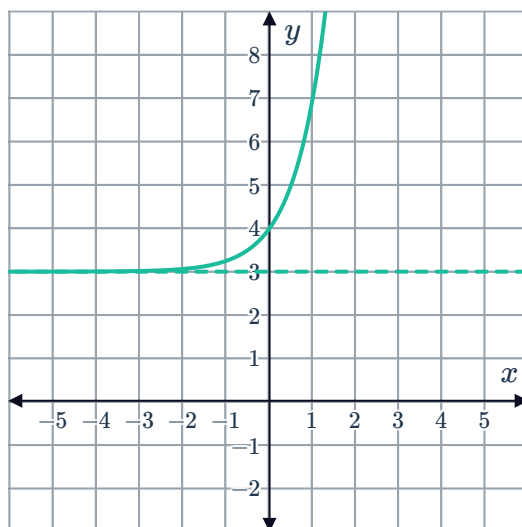
b

i 4

iii $y > 3$

ii All real x .

iv



12



a

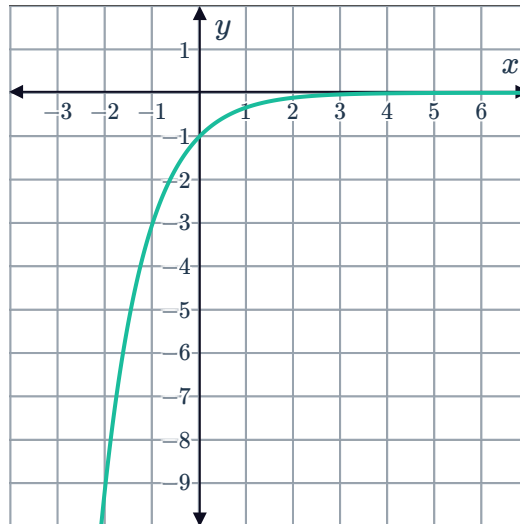
i -1

ii

x	-3	-2	-1	0	1	2	3
y	-27	-9	-3	-1	$-\frac{1}{3}$	$-\frac{1}{9}$	$-\frac{1}{27}$

iii $y = 0$

iv



v Increasing

b

i 1

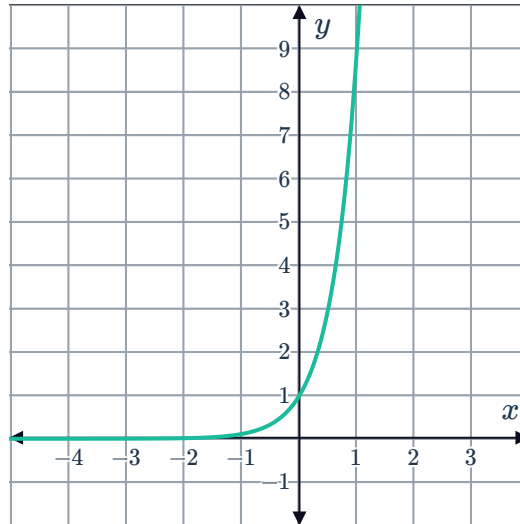


ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{729}$	$\frac{1}{81}$	$\frac{1}{9}$	1	9	81	729

iii $y = 0$

iv



v Increasing

c

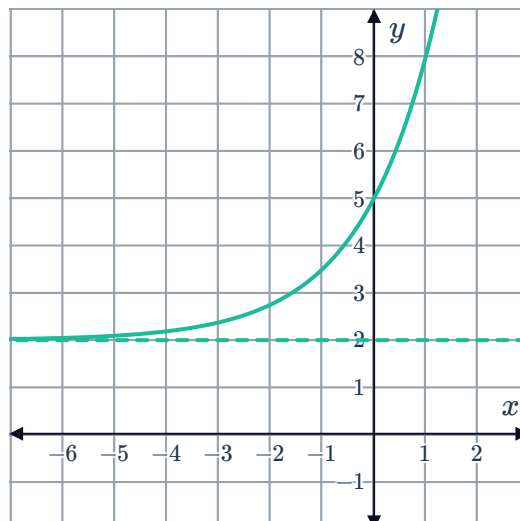
i 5

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{19}{8}$	$\frac{11}{4}$	$\frac{7}{2}$	5	8	14	26

iii $y = 2$

iv



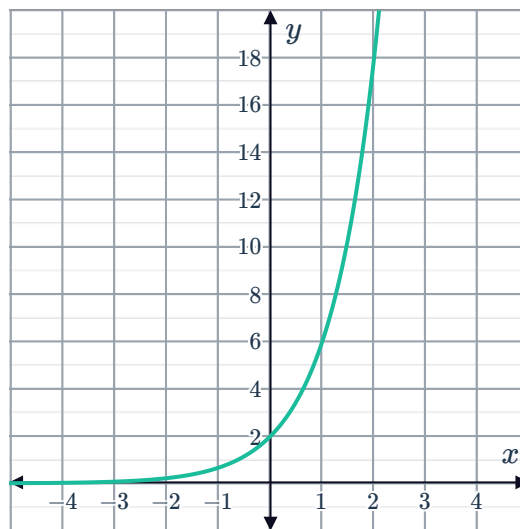


13

- a 0 b 6 c 0 d $y = 6$
 e All real x . f $y > 6$

14

- a 2
 b No. For $a > 0$ the expression a^x is greater than zero for all real x . Hence, $2 \times (a^x)$ will be greater than or equal to zero.
 c All real x .
 d $y > 0$
 e ∞
 f



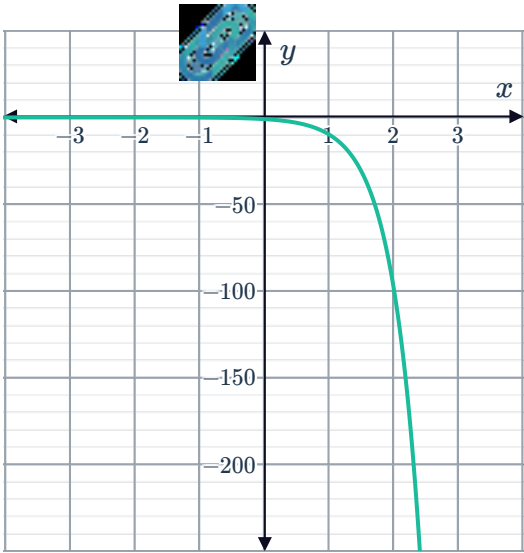
15

a

x	-2	-1	0	1	2	3
y	$-\frac{1}{100}$	$-\frac{1}{10}$	-1	-10	-100	-1000

- b Yes. For any value of x , 10^x is always positive. Therefore, -10^x is always negative.

c



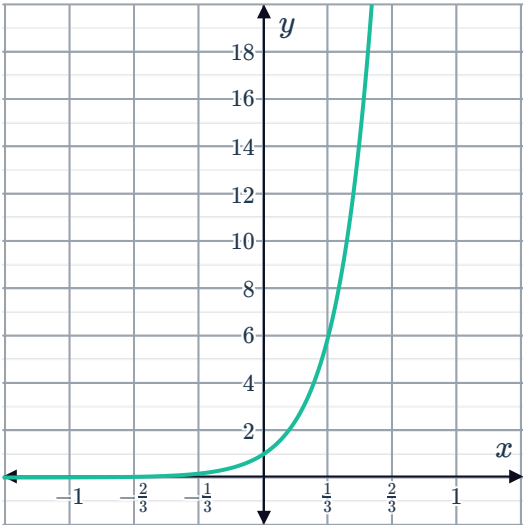
16

a $\frac{1}{3}$

b

- i No ii No iii No iv Yes

c



17

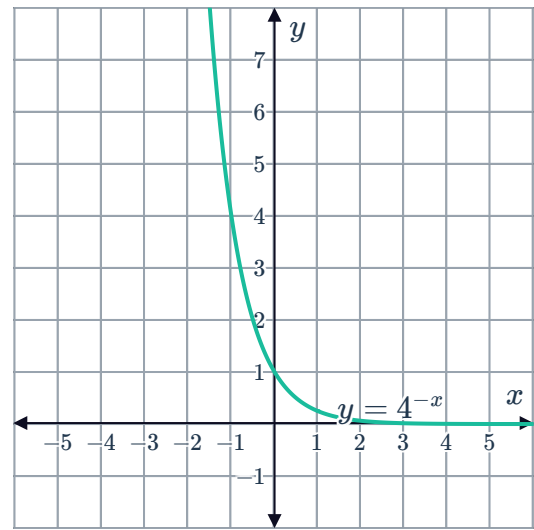
- a $y = 0$ b $y = 2$ c 1 d All real x .
e $y < 2$

18

a Reflection about the y -axis.



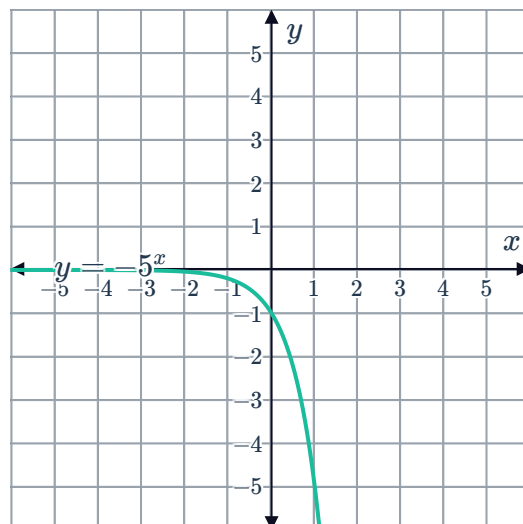
b



19

a A reflection about the x -axis

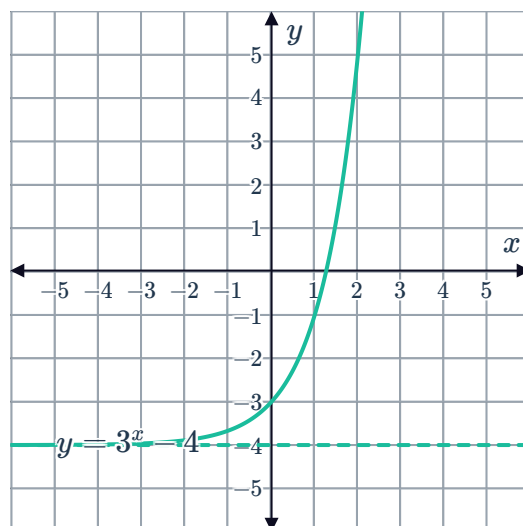
b



20

a A vertical translation by 4 units downwards.

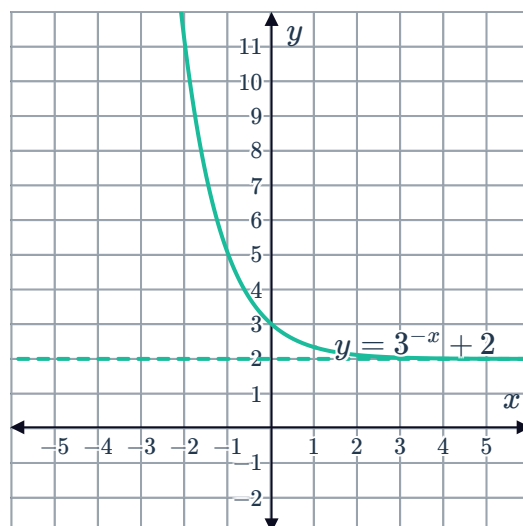
b



21

a A vertical translation by 2 units upwards.

b





Function from a graph

22

a No

b Yes

c Yes

d No

23 $g(x) = 9^x - 2$

24

a $y = 2^x$

b $y = 7^x$

c $y = 16^x$

d $y = 30^x$

e $y = (0.2)^x$

f $y = (0.7)^x$

25

a

i $y = 3$

ii $y = 7^x + 3$

b

i $y = -2$

ii $y = 7^x - 2$

c

i $y = 3$

ii $y = (0.5)^x + 3$

d

i $y = -6$

ii $y = \left(\frac{1}{3}\right)^x - 6$

26

a $y = 3 \times 2^x$

b $y = 10 \times 3^x$

c $y = 80 \times 0.25^x$

d $y = 90 \times \left(\frac{1}{3}\right)^x$

e $y = 5 \times 2^x$

f $y = -2 \times 3^x$